# Joseph Li

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## SKILLS

**Languages:** Python, C, C++, CUDA, Go, SQL

**Infrastructure:** Linux, Docker, Kubernetes, Git, GitHub Actions, CI/CD, QEMU, PostgreSQL, Redis

**ML & Systems:** PyTorch, vLLM, LoRA, Mixture-of-Experts, Quantization, Roofline Analysis, LLM Evaluation, Operating Systems, Virtual Memory, Observability, NumPy, Pandas

## PROFESSIONAL EXPERIENCE

**CORTEX AI** **Menlo Park, CA**

***AI Engineer Intern*** ***Jun. 2026 - Aug. 2026***

- Designed and launched an LLM API gateway in **Go**, **PostgreSQL**, and **Redis** that routes OpenAI, Anthropic, and **AWS Bedrock** through one endpoint for a 2,000-seat customer and meters $10K in monthly model spend
- Built a pre-release test suite with **GitHub Actions**, expanding coverage from 5 smoke tests to 159 tasks so tool regressions were caught before release
- Established live monitoring in **Langfuse**, adopted by all 9 engineers, tracking per-session latency and failures and routing each failure to the engineer responsible
- Reduced wasted model spend by stopping runaway agent loops and using an **LLM judge** to flag unnecessary tool calls

**YOYO MARKETS INC.** **New York, NY (Remote)**

***AI Engineer*** ***Aug. 2025 - Dec. 2025***

- Automated manual research and editorial work with a **Python** and LLM generation pipeline, producing **50+ markets per week at 5x the hand-curated rate**
- Developed an **asynchronous Python** scraper tracking 150+ high-signal Twitter accounts in real time, feeding that pipeline **3x the signal volume** of public trend aggregators
- Devised a ranking step that uses pairwise LLM comparisons and Elo ratings to screen 200+ candidate markets per cycle and send the strongest to founder review
- Shipped a **Next.js** review dashboard on **Vercel** that founders use daily, cutting **publishing turnaround from 30 minutes to under 5 minutes**

## RESEARCH AND PROJECTS

**EFFICIENT LLMS |** *PyTorch, CUDA, LoRA* **Jan. 2026 - May 2026 | USC**

- Converted **Llama 3.2-1B** into a **Mixture-of-Experts** model with **LoRA** adapters and kept held-out quality level with the original, perplexity **9.60 versus 9.66**
- Quantized weights from **FP16** to **INT4**, cutting model size **61%** while keeping generated answers coherent

**MATH REASONING FINE-TUNE |** *Python, LoRA* **Aug. 2025 - Dec. 2025 | USC**

- Trained **Qwen2.5-1.5B** with **LoRA** on 1,000 grade-school math problems to write its reasoning before the answer
- Rebuilt the grader until a correct final answer **scored consistently**, then implemented a **GRPO** trainer that rewards a correct, clearly reasoned solution among 4 candidates

**WEENIX (A UNIX-LIKE OS KERNEL) |** *C, x86, QEMU* **Aug. 2025 - Dec. 2025 | USC**

- Constructed a Unix-like kernel in **C**, with process scheduling, a file system, and virtual memory
- Integrated the three parts in **QEMU** so processes could be scheduled, files created and removed, and a copied process shared memory until one side wrote to it

## EDUCATION

**UNIVERSITY OF SOUTHERN CALIFORNIA** **Los Angeles, CA**

*M.S. Computer Science | GPA: 3.6/4.0* *Sep. 2024 - May 2026*

Relevant Coursework: ML Accelerators, Operating Systems

**UNIVERSITY OF MICHIGAN, ANN ARBOR** **Ann Arbor, MI**

*B.S.E. Data Science | GPA: 3.7/4.0* *Sep. 2020 - May 2024*